

Simone D'Ambrogio

DPHIL CANDIDATE

Department of Experimental Psychology, University of Oxford

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Current Position

Postdoctoral Researcher

DECISION AND ACTION LABORATORY, UNIVERSITY OF OXFORD

Oxford, UK

2026 – Present

Education

University of Oxford

DPHIL IN EXPERIMENTAL PSYCHOLOGY

Dissertation: *Learning Efficiently in Uncertain and Structured Worlds*

Advisors: Prof. Matthew F. S. Rushworth, Prof. Laurence T. Hunt

Examiners: Prof. Thomas Akam, Prof. Nathaniel D. Daw

Oxford, UK

2021 – 2026

University of Pennsylvania

VISITING GRADUATE SCHOLAR, WHARTON NEUROSCIENCE INITIATIVE

Philadelphia, PA

2020

University of Padua

M.SC. IN APPLIED COGNITIVE PSYCHOLOGY (CUM LAUDE)

Padua, Italy

2018 – 2020

Leiden University

ERASMUS+ PROGRAMME

Leiden, Netherlands

2018

University of Padua

B.SC. IN PSYCHOLOGY (FIRST CLASS HONOURS)

Padua, Italy

2014 – 2017

Research Interests

Decision-making · Computational modeling of cognition · Mechanistic Interpretability · Bayesian Inference · Generalization and Abstraction · Reinforcement learning

Publications

D'Ambrogio, S., Grohn, J., Khalighinejad, N., Mattar, M., Hunt, L., & Rushworth, M. F. S. (2026). *Interpretable abstractions of artificial neural networks predict behavior and neural activity during human information gathering*. *Nature Neuroscience*. <https://doi.org/10.1038/s41593-026-02342-9>

D'Ambrogio, S., Fei, Z., Platt, M., & Feng, S. (2026). *From pixels to preferences: Visual salience shapes economic choices beyond gaze alone*. *Nature Communications (Under Review)*.

Miyamoto*, K., **D'Ambrogio***, S., Harbison, C., Eichert, N., Schüffegen, U., Emberton, A., Sallet, J., Mars, R. B., Khalighinejad, N., & Rushworth, M. F. (2026). *Brain activity, disruption and connectivity comparisons identify origins of human metacognition in other primates* [* Equal contribution]. *Nature Human Behaviour*, 1–22. <https://doi.org/10.1038/s41562-026-02473-w>

Priolo*, G., Stablum*, F., Vacondio*, M., **D'Ambrogio, S.**, Caserotti, M., et al. (2026). *The robustness of mental accounting across 21 countries* [* Equal contribution]. *Journal of Consumer Research (In Press)*. <https://github.com/simonedambrogio/GlobalMentalAccounting>

Pan, D., **D'Ambrogio, S.**, Kingston, N., Rascu, M., Sankhe, P., Luo, S., Klein-Flügge, M. C., Mahmoodi, A., & Rushworth, M. F. S. (2025). *Construction and disruption of cognitive graphs in the human hippocampus*. *Nature Human Behaviour (Under Review)*, 2025.08.19.664920. <https://doi.org/10.1101/2025.08.19.664920>

D'Ambrogio, S., Werksman, N., Platt, M. L., & Johnson, E. N. (2023). *How celebrity status and gaze direction in ads drive visual attention to shape consumer decisions*. *Psychology & Marketing*, 40(4), 723–734. <https://doi.org/10.1002/mar.21772>

Ruggeri, K., Panin, A., ..., **D'Ambrogio, S.**, ..., Zupan, Z., & García-Garzon, E. (2022). *The globalizability of temporal discounting*. *Nature Human Behaviour*, 6(10), 1386–1397. <https://doi.org/10.1038/s41562-022-01392-w>

Research Experience

Decision and Action Laboratory, University of Oxford

Computational modeling, 7T fMRI, and non-invasive brain stimulation (TUS) to study information sampling, metacognition, and generalization in humans and macaques.

Oxford, UK

2021 – 2026

Platt Lab, Wharton Neuroscience Initiative, University of Pennsylvania

Eye-tracking, pupillometry, and drift-diffusion modeling of attention in consumer decision-making.

Philadelphia, PA

2020

Junior Researcher Programme

Computational modeling of decision-making and intertemporal choice.

Cambridge, UK

2020

Judgment and Decision-Making Lab, University of Padua

Behavioral experiments and modeling of judgment and decision-making for individual and societal well-being.

Padua, Italy

2018 – 2020

Psicostat Lab, University of Padua

Development of statistical methods for behavioral science.

Padua, Italy

2017 – 2020

Awards & Honours

Excellence Award

For research contributions beyond expected grade level. Funded by the University of Oxford
£ 1,044

UK

2025

PhD Scholarship in Psychological Sciences

Funded by the University of Padua
€ 45,343

Italy

2020

Erasmus+ Scholarship, School of Psychology

Funded by the University of Padua
€ 2,800

Italy

2018

Talks & Conference Presentations

Mathematics Of Neuroscience and AI (🎥 Talk)

Hybrid artificial neural network modelling predicts behaviour and neural activity in a solution for Buridan's ass.

Rome, Italy

2024

Foraging and Information Seeking Conference (📄 Poster)

Discovery of Cognitive Strategies for Information Sampling with Deep Cognitive Modelling and Investigation of their Neural Basis

Lyon, France

2024

Interdisciplinary Symposium on Decision Neuroscience (Poster)

Visuospatial working memory modulates the influence of visual attention on binary choice

Online

2021

Wharton Neuroscience Initiative Annual Meeting (Talk)

The effect of celebrity and visual attention on value-based decisions

Online

2021

Society for Neuroeconomics (Poster)

The Effect of Celebrity and Visual Attention on Value-Based Decisions

Online

2020

Italian Ministry of Economic Development – Division V (Talk)

Lessons from Decision Neuroscience to Inform Policy Making

Online

2019

Junior Researcher Programme Conference (Poster)

The Attentional Drift Diffusion Model with Temporal Information

Siena, Italy

2019

Psicostat Meeting (🌐 Talk)

Estimation of the attentional drift diffusion model parameters via Random Utility

Padua, Italy

2019

Teaching

Neuroanatomy, University of Oxford

Undergraduate tutorial teaching for Prof. Matthew F. S. Rushworth.

Oxford, UK

2022

Summer Schools

Computational Neuroscience, Neuromatch Academy

Project on fMRI decoding analysis supervised by Prof. Samuel J. Gershman

Online

2021

Psychology and Behavioral Sciences, JSchool

Studying how to improve real-world decision making

Siena, Italy

2020

Technical Skills & Methods

Computational Modeling

Artificial neural networks –
Symbolic regression –
Reinforcement learning –
Cognitive modelling – Bayesian
statistics

Neuroscience Methods

Ultra-high field fMRI (7T) –
Transcranial ultrasound
stimulation (TUS) – Eye-tracking
– Pupillometry – RSA – FSL

Programming & Infrastructure

Python – Julia – R – JavaScript –
HPC / SLURM – Git – $\text{\LaTeX}$ –
Quarto

References

Prof. Matthew F. S. Rushworth

Department of Experimental Psychology
University of Oxford
matthew.rushworth@psy.ox.ac.uk

Prof. Laurence T. Hunt

Department of Experimental Psychology
University of Oxford
laurence.hunt@psych.ox.ac.uk

Prof. Michael L. Platt

Department of Neuroscience
University of Pennsylvania
mplatt@pennmedicine.upenn.edu